

[Assignment VIZ]

Problem 1: Transpose of downsampling. Consider the downsampling operator $\mathcal{T}: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{(m/2) \times (n/2)}$, defined as the average pool with a 2×2 kernel and stride 2. For the sake of simplicity, assume m and n are even. Describe the action of $\mathcal{T}^\top$. More specifically, describe how to compute $\mathcal{T}^\top(Y)$ for any $Y \in \mathbb{R}^{(m/2) \times (n/2)}$.

Clarification. The downsampling operator $\mathcal{T}$ is a linear operator (why?). Therefore, $\mathcal{T}$ has a matrix representation $A \in \mathbb{R}^{(mn/4) \times (mn)}$ such that

$$\mathcal{T}(X) = (A(X.\text{reshape}(mn))).\text{reshape}(m/2, n/2)$$

for all $X \in \mathbb{R}^{m \times n}$. The adjoint $\mathcal{T}^\top$ has two equivalent definitions. One definition is

$$\mathcal{T}^\top(Y) = (A^\top(Y.\text{reshape}(mn/4))).\text{reshape}(m, n)$$

for all $Y \in \mathbb{R}^{(m/2) \times (n/2)}$. Another is

$$\sum_{i=1}^{m/2} \sum_{j=1}^{n/2} Y_{ij} (\mathcal{T}(X))_{ij} = \sum_{i=1}^m \sum_{j=1}^n (\mathcal{T}^\top(Y))_{ij} (X)_{ij}$$

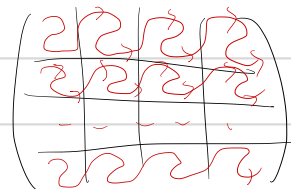
for all $X \in \mathbb{R}^{m \times n}$ and $Y \in \mathbb{R}^{(m/2) \times (n/2)}$.

Hint. To spoil the suspense, $\mathcal{T}^\top$ is a constant times the nearest neighbor upsampling. Explain why in your answer.

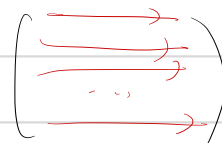
(손쉬운 설명)

$$\mathbb{R}^{m \times n}: \{e_1, e_2, e_2, e_2, e_3, e_4, e_3, e_4, \dots\}$$

$$\dots\} = 0$$



$$\mathbb{R}^{m/2 \times n/2}: \{e_1, e_2, \dots\} = e$$



$$\circ \quad \mathcal{T}; \text{ (nearest map?) } \Rightarrow \mathcal{T}(C(N)) = \mathcal{T}(C(N)), \quad \mathcal{T}(C(M)) = C(\mathcal{T}(M)) \text{ . obvious}$$

$\Rightarrow$ 손쉬운 설명

$$[\mathcal{T}]_C^C = \begin{pmatrix} \begin{matrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} \end{pmatrix}$$

$$\in \mathbb{R}^{mn/4 \times mn}$$

, B, C ; orthonormal basis

$$\Rightarrow [\mathcal{T}^\top]_B^C = \begin{pmatrix} \begin{matrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{matrix} \\ \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} 0 & 0 & 0 & 0 \end{matrix} \end{pmatrix}$$

$$\in \mathbb{R}^{mn \times mn/4}$$

$$\Rightarrow [\mathcal{T}^\top]_B^C = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1, n/2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m/2, 1} & x_{m/2, 2} & \dots & x_{m/2, n/2} \end{pmatrix}$$

$$= \begin{pmatrix} x_{11} & x_{11} & \dots & x_{11} \\ x_{12} & x_{12} & \dots & x_{12} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m/2, 1} & x_{m/2, 1} & \dots & x_{m/2, 1} \\ x_{m/2, 2} & x_{m/2, 2} & \dots & x_{m/2, 2} \end{pmatrix}$$

($\frac{1}{4} \times$ nearest neighbour upsampling)

with ... appropriately filled in.

은 구멍이 5(바늘), 허렁이는 6(줄)씩 3(바늘) 된다.

$$\Leftarrow \underbrace{\begin{pmatrix} A & & & \\ & \textcircled{0} & & \\ & & \ddots & \\ & & & \textcircled{0} \end{pmatrix}}_C \cdot \underbrace{\begin{pmatrix} \textcircled{0} & & & \\ & \ddots & & \\ & & \textcircled{0} & \\ & & & A \end{pmatrix}}_C \cdot \underbrace{\begin{pmatrix} \textcircled{0} & & & \\ & \ddots & & \\ & & \textcircled{0} & \\ & & & A \end{pmatrix}}_C \cdot A \Leftarrow \checkmark$$

은 100%인 것이다.

Problem 3: The true softmax. Define

$$\nu_\beta(x) = \frac{1}{\beta} \log \sum_{i=1}^n \exp(\beta x_i).$$

Clearly, $\nu_\beta: \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable. Show that

(a) $\nu_\beta(x) \rightarrow \max\{x_1, \dots, x_n\}$ as $\beta \rightarrow \infty$.

최대값을 갖는 x_i $x_i = \max\{x_1, \dots, x_n\}$ $\rightarrow$ $\lim_{\beta \rightarrow \infty} \nu_\beta(x) = x_i$

$$\nu_\beta(x) = \frac{1}{\beta} \log (e^{\beta x_1} (1 + e^{\beta(x_2 - x_1)} + \dots + e^{\beta(x_n - x_1)}))$$

$$= x_1 + \frac{1}{\beta} \log (1 + e^{\beta(x_2 - x_1)} + \dots + e^{\beta(x_n - x_1)})$$

$$\beta \rightarrow \infty; 1 \leq 1 + e^{\beta(x_2 - x_1)} + \dots + e^{\beta(x_n - x_1)} \leq n$$

$$\nu_\beta(x) \rightarrow x_1$$

$\nearrow$ if $x_1 = x_2, \dots$ $\searrow$

(b) $\nabla \nu_1 = \mu$, where μ is the softmax function.

$$\nu(x) = \log \sum_{i=1}^n \exp(x_i)$$

$$\Rightarrow \frac{\partial \nu}{\partial x_i} = \frac{\exp(x_i)}{\sum_{j=1}^n \exp(x_j)} \Rightarrow \nabla \nu = \frac{1}{\sum_{j=1}^n \exp(x_j)} (\exp(x_1), \dots, \exp(x_n)) = \text{softmax}(x) //$$

(c) If $i_{\max} = \operatorname{argmax}_{1 \leq i \leq n} x_i$ is uniquely defined, then $\nabla \nu_\beta(x) \rightarrow e_{i_{\max}}$ as $\beta \rightarrow \infty$, where $\{e_1, \dots, e_n\}$ is the standard basis of $\mathbb{R}^n$.

$$\frac{\partial \nu}{\partial x_i} = \frac{1}{n} \frac{\exp(\beta x_i)}{\sum_{j=1}^n \exp(\beta x_j)} = \frac{\exp(\beta x_i)}{\sum_{j=1}^n \exp(\beta x_j)} = \frac{\exp(\beta x_i - \beta x_{i_{\max}})}{\sum_{j=1}^n \exp(\beta x_j - \beta x_{i_{\max}})}$$

$$\sum_{j=1}^n \exp(\beta x_j - \beta x_{i_{\max}}) = 1 + \sum_{j \neq i_{\max}} \exp(\beta x_j - \beta x_{i_{\max}}) < 0 \text{ c' } i_{\max} = \operatorname{argmax}_{1 \leq i \leq n} x_i \rightarrow 1 \text{ if } \beta \rightarrow \infty$$

$$\text{if } i = i_{\max} \quad \exp(\beta x_i - \beta x_{i_{\max}}) = 1$$

$$\text{else} \quad \exp(\beta x_i - \beta x_{i_{\max}}) \rightarrow 0 \text{ if } \beta \rightarrow \infty$$

$$\Rightarrow \nabla \nu_\beta(x) \rightarrow e_{i_{\max}} //$$

Problem 4: Generalized inverse transform sampling. Let $F: \mathbb{R} \rightarrow [0, 1]$ be the CDF of a random variable, and let $U \sim \text{Uniform}([0, 1])$. If F is strictly increasing and therefore invertible, then $F^{-1}(U)$ is a random variable with CDF F , because

$$\mathbb{P}(F^{-1}(U) \leq t) = \mathbb{P}(U \leq F(t)) = F(t).$$

When F is not necessarily invertible, the *generalized inverse* of F is $G: (0, 1) \rightarrow \mathbb{R}$ with

$$G(u) = \inf\{x \in \mathbb{R} \mid u \leq F(x)\}.$$

Show $G(U)$ with is a random variable with CDF F .

Hint. Use the fact that F is right-continuous, i.e., $\lim_{h \rightarrow 0^+} F(x+h) = F(x)$ for all $x \in \mathbb{R}$, and that $\lim_{x \rightarrow -\infty} F(x) = 0$.

$t \in \mathbb{R}$ and $\{u: G(u) \leq t\} = \{u: u \leq F(t)\}$ are both sets of u .

$$\textcircled{1} \quad G(u) \leq t \Rightarrow u \leq F(t)$$

$$\inf\{x \in \mathbb{R} : u \leq F(x)\} \leq t \Rightarrow A = \{x \in \mathbb{R} : u \leq F(x)\} \text{ is non-empty} \quad A \neq \emptyset, \quad \forall u > 0 \quad u \leq F(A) \text{ (i.e., } F \text{ is monotone)}$$

$$\Rightarrow \exists \epsilon > 0 : u \leq F(t) \leq F(t) \quad (\text{by right-continuity of CDF})$$

$$\textcircled{2} \quad u \leq F(t) \Rightarrow G(u) \leq t$$

$$t \in \{x \in \mathbb{R} : u \leq F(x)\} \Rightarrow G(u) = \inf\{x \in \mathbb{R} : u \leq F(x)\} \leq t \quad \square$$

To prove this property
requires some knowledge of
probability measure theory.

Problem 5: f -divergence. Let X and Y be two continuous random variables with densities p_X and p_Y . The f -divergence of X from Y is defined as

$$D_f(X\|Y) = \int f\left(\frac{p_X(x)}{p_Y(x)}\right) p_Y(x) dx,$$

where f is a convex function such that $f(1) = 0$.

(a) Show that $D_f(X\|Y) \geq 0$.

Jensen's inequality $D_f(X\|Y) = \int f\left(\frac{p_X(x)}{p_Y(x)}\right) p_Y(x) dx \geq f\left(\int \frac{p_X(x)}{p_Y(x)} p_Y(x) dx\right) = f\left(\int p_X(x) dx\right) = f(1) = 0$ (f: convex)

(b) Show that $f = -\log t$ and $f = t \log t$ correspond to KL divergence.

$$f = -\log t \Rightarrow D_f(X\|Y) = \int -\log\left(\frac{p_X(x)}{p_Y(x)}\right) p_Y(x) dx = \int p_Y(x) \log\left(\frac{p_Y(x)}{p_X(x)}\right) dx = D_{KL}(Y\|X),$$

$$f = t \log t \Rightarrow D_f(X\|Y) = \int \frac{p_X(x)}{p_Y(x)} \cdot \log\left(\frac{p_X(x)}{p_Y(x)}\right) p_Y(x) dx = \int p_X(x) \log\left(\frac{p_X(x)}{p_Y(x)}\right) dx = D_{KL}(X\|Y),$$

Problem 6: Backprop with convolutions. Consider 1D convolutions with single input and output channels, stride 1, and padding 0. Let $w_1, \dots, w_L$ be convolutional filters with sizes $f_1, \dots, f_L$. Let $A_{w_\ell} \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$, where $n_\ell = n_{\ell-1} - f_\ell + 1$, be the matrix representing convolution with w_ℓ , i.e., multiplication by A_{w_ℓ} is equivalent to convolution with w_ℓ , for $\ell = 1, \dots, L$. Let $\sigma: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable activation function. Consider the convolutional neural network

$$\begin{aligned} y_L &= A_{w_L} y_{L-1} + b_L \mathbf{1}_{n_L} \\ y_{L-1} &= \sigma(A_{w_{L-1}} y_{L-2} + b_{L-1} \mathbf{1}_{n_{L-1}}) \\ &\vdots \\ y_2 &= \sigma(A_{w_2} y_1 + b_2 \mathbf{1}_{n_2}) \\ y_1 &= \sigma(A_{w_1} x + b_1 \mathbf{1}_{n_1}), \end{aligned}$$

where $x \in \mathbb{R}^{n_0}$, $b_\ell \in \mathbb{R}$, $\mathbf{1}_{n_\ell} \in \mathbb{R}^{n_\ell}$ is the vector with all entries being 1, and $n_L = 1$. For notational convenience, define $y_0 = x$.

(a) Define

$$v_L = 1, \quad v_\ell = \frac{\partial y_L}{\partial y_\ell} \text{diag}(\sigma'(A_{w_\ell} y_{\ell-1} + b_\ell \mathbf{1}_{n_\ell})) \quad \text{for } \ell = 1, \dots, L-1.$$

Let $C_{v_\ell^\top}$ be the 1D convolutional operator defined by interpreting $v_\ell^\top \in \mathbb{R}^{n_\ell}$ as a convolutional filter for $\ell = 1, \dots, L$. Show that

$$\begin{aligned} \frac{\partial y_L}{\partial y_{L-1}} &= A_{w_L}, & \frac{\partial y_\ell}{\partial y_{\ell-1}} &= \text{diag}(\sigma'(A_{w_\ell} y_{\ell-1} + b_\ell \mathbf{1}_{n_\ell})) A_{w_\ell} \quad \text{for } \ell = 2, \dots, L-1 \\ \frac{\partial y_L}{\partial w_\ell} &= (C_{v_\ell^\top} y_{\ell-1})^\top & \text{for } \ell = 1, \dots, L \\ \frac{\partial y_L}{\partial b_\ell} &= v_\ell \mathbf{1}_{n_\ell} & \text{for } \ell = 1, \dots, L. \end{aligned}$$

1D convolutional filter의 공배는 2배

(b) As discussed in homework 1, forming the full matrix A_{w_ℓ} is wasteful and should be avoided. Describe how matrix-vector or vector-matrix products with respect to A_{w_ℓ} or $A_{w_\ell}^\top$ should be used in the forward pass and backpropagation.

이전 forward mode의 경우 y_ℓ 의 값이 0인 경우 무시됨.

이전 backward mode의 경우 $\frac{\partial y_L}{\partial y_\ell}$ 의 값을 A_{w_ℓ} 로 곱함

1D convolutional filter가 일정한 크기이고, $\frac{\partial y_L}{\partial b_\ell}$ 은 $\frac{\partial y_L}{\partial y_\ell} \text{diag}(\sigma'(y_\ell))$

$$\textcircled{1} y_L = \sum_j (A_{w_L})_{1,j} y_{L-1,j} + b_L \Rightarrow \frac{\partial y_L}{\partial y_{L-1,j}} = (A_{w_L})_{1,j}, \quad \frac{\partial y_L}{\partial y_{L-1}} = A_{w_L}$$

임의의 j 에 대한 $\frac{\partial y_L}{\partial y_{L-1,j}}$ 의 forward mode의 계산은 A_{w_L} 의 j 번째 열을 곱하는 것과 동일함. $\frac{\partial y_L}{\partial y_{L-1}}$ 은 $C_{v_L^\top}$ 과 같은 1D

$$\textcircled{2} y_{L-1} = \sigma(\sum_j (A_{w_{L-1}})_{1,j} y_{L-2,j} + b_{L-1}) \Rightarrow \frac{\partial y_{L-1}}{\partial y_{L-2,j}} = \sigma'(A_{w_{L-1}} y_{L-2} + b_{L-1}) \Rightarrow \frac{\partial y_L}{\partial y_{L-2}} = \text{diag}(\sigma') A_{w_{L-1}}$$

convolutional filter의 forward

pass의 경우 y_{L-1} 의 모든 요소를 곱함.

따라서 많은 연산량을 줄일 수 있음.

$$\textcircled{3} \text{if } \frac{\partial y_L}{\partial b_L} = \sigma' \Rightarrow \frac{\partial y_L}{\partial b_L} = \text{diag}(\sigma') \cdot \mathbf{1}_{n_L}, \quad \frac{\partial y_L}{\partial b_L} = \frac{\partial y_L}{\partial y_L} \frac{\partial y_L}{\partial b_L} = v_L \mathbf{1}_{n_L}.$$

$$\textcircled{4} \frac{\partial y_L}{\partial w_\ell} = \frac{\partial y_L}{\partial y_\ell} \frac{\partial y_\ell}{\partial w_\ell} = \frac{\partial y_L}{\partial y_\ell} \frac{\partial y_\ell}{\partial y_{\ell-1}} \frac{\partial y_{\ell-1}}{\partial w_\ell} = \frac{\partial y_L}{\partial y_\ell} \text{diag}(\sigma') \frac{\partial y_{\ell-1}}{\partial w_\ell} = v_\ell \frac{\partial y_{\ell-1}}{\partial w_\ell} \quad \text{이전}$$

$$\left(\frac{\partial y_L}{\partial w_\ell} \right)_k = \frac{\partial (A_{w_\ell} y_{\ell-1} + b_\ell \mathbf{1}_{n_\ell})_k}{\partial (w_\ell)_j} = \frac{\partial (\sum_{m=1}^{f_\ell} (w_\ell)_{m,j} (y_{\ell-1})_{k-m+1} + b_\ell)}{\partial (w_\ell)_j} = (y_{\ell-1})_{k-j+1} \quad (1 \leq j \leq f_\ell) \text{ 이고}$$

$$\left(\frac{\partial y_L}{\partial w_\ell} \right)_j = \sum_{k=1}^{n_\ell} (v_\ell^\top)_k (y_{\ell-1})_{k-j+1} \quad (1 \leq j \leq f_\ell) \text{ 이고, } (C_{v_\ell^\top} y_{\ell-1})_j = \sum_{k=1}^{n_\ell} (y_{\ell-1})_{j+k-1} (v_\ell^\top)_k \quad (1 \leq j \leq f_\ell) \text{ 이고}$$

$$\frac{\partial y_L}{\partial w_\ell} = (C_{v_\ell^\top} y_{\ell-1})^\top$$